

The square set, $u' = 0$; and x -formulæ.

37. We use the square-set, in the first instance by writing therein $u' = 0$; the equations become

$$\begin{aligned} A^2u &= \alpha X + \beta Y, &= \omega''\mathfrak{A}(a-x), \\ B^2u &= \beta X + \alpha Y, &= \omega''\mathfrak{B}(b-x), \\ C^2u &= \alpha X - \beta Y, &= \omega''\mathfrak{C}(c-x), \\ D^2u &= \beta X - \alpha Y, &= \omega''\mathfrak{D}(d-x), \end{aligned}$$

viz. the equations without their last members show that there exist functions ω'' and x , linear functions of X and Y , such that $\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{D}$, $\mathfrak{A}a, \mathfrak{B}b, \mathfrak{C}c, \mathfrak{D}d$, being constants, the squared functions may be assumed equal to $\mathfrak{A}\omega'' \cdot \omega''x$, &c., that is, $\omega''\mathfrak{A}(a-x)$, &c., respectively; the squared functions are then proportional to the values $\mathfrak{A}(a-x)$, &c.

To show the meaning of the factor ω'' , observe that, from any two of the equations, for instance from the first and second, we have an equation without ω ,

$$A^2u - B^2u = \mathfrak{A}(a-x) - \mathfrak{B}(b-x);$$

and using this to determine x , and then substituting in $\omega'' = A^2u \div \mathfrak{A}(a-x)$, we find

$$\omega'' = \frac{\mathfrak{A}A^2u - \mathfrak{B}B^2u}{(a-b)\mathfrak{A}\mathfrak{B}},$$

where the numerator is a function not in anywise more important than any other linear function of A^2u and B^2u .

38. The function Du vanishes for $u = 0$, and we may assume that the corresponding value of x is $= d$. Writing in the other equations $u = 0$, they become

$$\begin{aligned} A^20 &= \alpha_0^2 + \beta_0^2 = \omega_0''\mathfrak{A}(a-d), \\ B^20 &= 2\alpha_0\beta_0 = \omega_0''\mathfrak{B}(b-d), \\ C^20 &= \alpha_0^2 - \beta_0^2 = \omega_0''\mathfrak{C}(c-d), \end{aligned}$$

where ω_0'' is what ω'' becomes on writing therein $x = d$. It is convenient to omit altogether these factors ω'' and ω_0'' ; it being understood that, without them, the equations denote not absolute equalities but only equalities of ratios: thus, without the ω_0'' , the last-mentioned equations would denote

$$A^20 : B^20 : C^20 = \alpha_0^2 + \beta_0^2 : 2\alpha_0\beta_0 : \alpha_0^2 - \beta_0^2 = \mathfrak{A}(a-d) : \mathfrak{B}(b-d) : \mathfrak{C}(c-d).$$

The quantities $\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{D}$ only present themselves in the products $\mathfrak{A}\omega''$, &c., and their absolute magnitudes are therefore essentially indeterminate; but regarding ω'' as

containing a constant factor of properly determined value, the absolute values of $\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{D}$ may be regarded as determinate, and this is accordingly done in the formulæ $\mathfrak{A}^2 = -agh$, &c., which follow.

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Relations between the constants.

39. The formulæ contain the differences of the quantities a, b, c, d ; denoting these differences

$$b - c, \quad c - a, \quad a - b, \quad a - d, \quad b - d, \quad c - d,$$

in the usual manner by

$$a, \quad b, \quad c, \quad f, \quad g, \quad h,$$

so that

$$-h + g - a = 0,$$

$$h - f - b = 0,$$

$$-g + f - c = 0,$$

$$a + b + c = 0,$$

and also

$$af + bg + ch = 0,$$

and then assuming the absolute value of one of the quantities $\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{D}$, we have the system of relations

$$\mathfrak{A}^2 = -agh, \quad \mathfrak{B}\mathfrak{C}\mathfrak{D} = \mathfrak{A}^2\mathfrak{D}f, \quad \mathfrak{A}bcf = -\mathfrak{B}\mathfrak{C}\mathfrak{D}, \quad \mathfrak{A}\mathfrak{B}\mathfrak{C}\mathfrak{D} = abc fgh,$$

$$\mathfrak{B}^2 = bhf, \quad \mathfrak{C}\mathfrak{D}\mathfrak{A} = -\mathfrak{B}^2\mathfrak{D}g, \quad \mathfrak{B}cag = \mathfrak{C}\mathfrak{A}\mathfrak{D},$$

$$\mathfrak{C}^2 = cfg, \quad \mathfrak{A}\mathfrak{B}\mathfrak{C} = -\mathfrak{C}^2\mathfrak{D}h, \quad \mathfrak{C}abh = \mathfrak{A}\mathfrak{B}\mathfrak{D},$$

$$\mathfrak{D}^2 = -abc, \quad \mathfrak{D}fgh = -\mathfrak{A}\mathfrak{B}\mathfrak{C},$$

$$0 = \mathfrak{B}^2 + b^2\mathfrak{C}^2 - f^2\mathfrak{D}^2 = bcf (af + bg + ch) = 0,$$

$$0 = -c^2\mathfrak{A}^2 + a^2\mathfrak{C}^2 - g^2\mathfrak{D}^2 = cag (\quad) = 0,$$

$$0 = -b^2\mathfrak{A}^2 + a^2\mathfrak{B}^2 - h^2\mathfrak{D}^2 = abh (\quad) = 0,$$

$$0 = -f^2\mathfrak{A}^2 + g^2\mathfrak{B}^2 + h^2\mathfrak{C}^2 = fgh (\quad) = 0.$$

It is to be remarked that, taking c, a, b, d in the order of decreasing magnitude, we have $-a, b, c, f, g, h$ all positive; hence $\mathfrak{A}^2, \mathfrak{B}^2, \mathfrak{C}^2, \mathfrak{D}^2$ all real; and taking as we may do, $\mathfrak{D}$ negative, then $\mathfrak{A}, \mathfrak{B}, \mathfrak{C}$ may be taken positive; that is, we have $-a, b, c, f, g, h, \mathfrak{A}, \mathfrak{B}, \mathfrak{C}, -\mathfrak{D}$ all of them positive.

40. We have

$$A^2 0 = \alpha_0^2 + \beta_0^2 = \mathfrak{A}f,$$

$$B^2 0 = 2\alpha_0\beta_0 = \mathfrak{B}g,$$

$$C^2 0 = \alpha_0^2 - \beta_0^2 = \mathfrak{C}h.$$

The foregoing equations

$$k = \frac{B^2 0}{A^2 0}, \quad k'^2 = \frac{C^2 0}{A^2 0}$$

give

$$k = \frac{\mathfrak{B}g}{\mathfrak{A}f}, \quad k'^2 = \frac{\mathfrak{C}h}{\mathfrak{A}f}.$$

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and we thence have

$$k^2 = \frac{bg}{-af}, \quad k'^2 = \frac{ch}{-af},$$

satisfying

$$k^2 + k'^2 = 1.$$

41. Observe further that, substituting for a, b, c, f, g, h their values, we have

$$\begin{aligned} \mathfrak{A}^2 &= c - b.b - d.c - d, &= c - a.d - b.b - c, \\ \mathfrak{B}^2 &= c - a.c - d.a - d, &= d - a.a - c.c - d, \\ \mathfrak{C}^2 &= a - b.a - d.b - d, &= -a - b.b - d.d - a, \\ \mathfrak{D}^2 &= c - b.c - a.a - b, &= -b - c.c - a.a - b, \end{aligned}$$

where in the first set of values all the differences are positive, but in the second set of values, we take the triads of $abcd$, in the cyclical order bcd, cda, dab, abc . There is in this last form an apparent want of symmetry as to the signs (viz. the order which might have been expected is $+-+-$), but taking the order of the letters to be $\mathfrak{C}, \mathfrak{A}, \mathfrak{B}, \mathfrak{D}$ and c, a, b, d , then the cyclical arrangement is

$$\begin{aligned} \mathfrak{C}^2 &= -b - d.d - a.a - b, \\ \mathfrak{A}^2 &= -d - c.c - b.b - d, \\ \mathfrak{B}^2 &= -c - a.a - d.d - c, \\ \mathfrak{D}^2 &= -a - b.b - c.c - a, \end{aligned}$$

where the four outside signs are all $-$. Observe that the triads of $abcd$, and $abdc$, are

$$bcd, \quad cda, \quad dab, \quad abc,$$

and

$$bdc, \quad dca, \quad cab, \quad abd.$$

where in the first and second columns the terms of the same column correspond to each other with a reversal of sign, whereas in the third and fourth columns the lower term of either column corresponds to the upper term of the other column, but without a reversal of sign.

The product-sets, $u \pm u'$; and u' indefinitely small, differential formulæ.

42. Coming now to the product-sets, these may be written

$$\begin{aligned} \frac{1}{2}\{C.A + A.C\} &= X_1 X'_1, & \frac{1}{2}\{C.A - A.C\} &= Y_1 Y'_1, \\ \frac{1}{2}\{D.B + B.D\} &= Y_1 X'_1, & \frac{1}{2}\{D.B - B.D\} &= X_1 Y'_1, \\ \frac{1}{2}\{B.A + A.B\} &= (P + Q)(P' + Q'), & \frac{1}{2}\{B.A - A.B\} &= (P - Q)(P' - Q'), \\ \frac{1}{2}\{D.C + C.D\} &= i(P - Q)(P' + Q'), & \frac{1}{2}\{D.C - C.D\} &= i(P + Q)(P' - Q'), \\ \frac{1}{2}\{D.A + A.D\} &= (P_1 - iQ_1)(P'_1 + iQ'_1), & \frac{1}{2}\{D.A - A.D\} &= (P_1 + iQ_1)(P'_1 - iQ'_1), \\ \frac{1}{2}\{B.C + C.B\} &= -i(P_1 - iQ_1)(P'_1 + iQ'_1), & \frac{1}{2}\{B.C - C.B\} &= -i(P_1 + iQ_1)(P'_1 - iQ'_1). \end{aligned}$$

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43. We can from each set form two fractions (each of them a function of $u + u'$ and $u - u'$), which are equal to one and the same function of u' only: for instance, from the first set we have two fractions, each $= \frac{Y'}{X'}$; putting in such equation $u = 0$, we obtain a new expression for the function of u' involving only the theta-functions Au' , &c., which new expression we may then substitute in the equations first obtained; we thus arrive at the six equations

$$\begin{aligned} \frac{C.A - A.C}{D.B + B.D} &= \frac{D.B - B.D}{C.A + A.C} = \frac{D'u' . B'u'}{C'u' . A'u'}, \\ \frac{B.A - A.B}{D.C + C.D} &= \frac{D.C - C.D}{B.A + A.B} = \frac{D'u' . C'u'}{B'u' . A'u'}, \\ \frac{B.C - C.B}{D.A + A.D} &= \frac{D.A - A.D}{B.C + C.B} = \frac{D'u' . A'u'}{B'u' . C'u'}, \end{aligned}$$

where observe that the expressions all vanish for $u' = 0$.

44. Taking herein u' indefinitely small, we obtain

$$\begin{aligned} \frac{Au . C'u - Cu . A'u}{Bu . Du} &= \frac{Bu . D'u - Du . B'u}{Cu . Au} = \frac{D'0 . B0}{C0 . A0} = -K \frac{D^2 0}{A^2 0}, \\ \frac{Au . B'u - Bu . A'u}{Cu . Du} &= \frac{Cu . D'u - Du . C'u}{Au . Bu} = \frac{D'0 . C0}{A0 . B0} = -K \frac{C^2 0}{A^2 0}, \\ \frac{Cu . B'u - Bu . C'u}{Au . Du} &= \frac{Au . D'u - Du . A'u}{Bu . Cu} = \frac{D'0 . A0}{B0 . C0} = -K, \end{aligned}$$

where the last column is added in order to introduce K in place of $D'0$.

45. These formulae in effect give the derivatives of the quotient-functions in terms of quotient-functions: for instance, one of the equations is

$$\frac{d}{du} \frac{Du}{Au} = -K \frac{Bu \ Cu}{Au \ Au},$$

substituting herein for the quotient-fractions their values in terms of x , this becomes

$$\frac{d}{du} \frac{1}{\mathfrak{A}} \sqrt{\frac{a-x}{d-x}} = -\frac{K}{\mathfrak{A}^2} \sqrt{\mathfrak{B}\mathfrak{C}} \sqrt{\frac{a-x}{d-x}} \cdot \frac{1}{\mathfrak{A}} \sqrt{\frac{a-x}{d-x}},$$

or the left-hand being

$$-\frac{1}{(a-x)^{\frac{1}{2}} \sqrt{d-x}} \frac{dx}{du},$$

this is

$$K \ du = \frac{\frac{1}{2} \sqrt{-af} \ dx}{\sqrt{a-x . b-x . c-x . d-x}},$$

where on the right-hand side it would be better to write $\sqrt{-af}$ in the numerator and $x-d$ in place of $d-x$ in the denominator.

Comparison with Jacobi's formulæ.

46. The comparison of the formulæ with Jacobi's formulæ gives

$$\begin{aligned}\operatorname{sn} Ku &= -\frac{1}{\sqrt{k}} Du \div Cu, &= \sqrt{\frac{h}{g}} \sqrt{\frac{d-x}{c-x}} \quad (\text{or better } \sqrt{\frac{-h}{g}} \sqrt{\frac{x-d}{c-x}}), \\ \operatorname{cn} Ku &= \sqrt{\frac{k'}{k}} Bu \div Cu, &= \sqrt{\frac{h}{g}} \sqrt{\frac{b-x}{c-x}}, \\ \operatorname{dn} Ku &= \sqrt{k'} Au \div Cu, &= \sqrt{\frac{h}{f}} \sqrt{\frac{a-x}{c-x}},\end{aligned}$$

where it will be recollected that

$$k^2 = \frac{bg}{-af}, \quad k'^2 = \frac{ch}{-af}.$$

It may be remarked that we seek to determine everything in terms of a, b, c, d . The formula just written down, $f^2 = bg + h - af$, gives k in terms of these quantities; and k, K being each given in terms of q , we have virtually K as a function of k , that is, of a, b, c, d : but it would not be easy from the expressions of k, K , each in terms of q , to deduce the actual expression

$$K = \int_0^{\pi/2} \frac{d\phi}{\sqrt{1 - k^2 \sin^2 \phi}}$$

of K as a function of k .

The square-set, $u \pm u'$.

47. Reverting to the square-set

$$\begin{aligned}A(u + u')A(u - u') &= XX' + YY', \\ B(u + u')B(u - u') &= YX' + XY', \\ C(u + u')C(u - u') &= XX' - YY', \\ D(u + u')D(u - u') &= -YX' + XY',\end{aligned}$$

if we first write herein $u' = 0$, and then $u = 0$, using the results to determine the values of X, Y, X', Y' we find

$$\begin{aligned}\alpha C^2 u - \beta D^2 u &= (\alpha^2 - \beta^2)X, & \alpha C^2 u' - \beta D^2 u' &= (\alpha^2 - \beta^2)X', \\ \beta C^2 u - \alpha D^2 u &= Y, & \beta C^2 u' - \alpha D^2 u' &= Y',\end{aligned}$$

and thence

$$\begin{aligned}(\alpha^2 - \beta^2)^2 XX' &= \alpha^2 C^2 u C^2 u' + \beta^2 D^2 u D^2 u' - \alpha\beta (C^2 u D^2 u' + D^2 u C^2 u'), \\ YY' &= \beta^2 \text{ " } + \alpha^2 \text{ " } - \alpha\beta \text{ " },\end{aligned}$$

whence

$$\begin{aligned}(\alpha^2 - \beta^2)^2 (XX' + YY') &= (\alpha^2 + \beta^2)(C^2 u C^2 u' + D^2 u D^2 u') - 2\alpha\beta (C^2 u D^2 u' + D^2 u C^2 u'), \\ (\alpha^2 - \beta^2)^2 (XX' - YY') &= (C^2 u C^2 u' - D^2 u D^2 u'),\end{aligned}$$

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where observe that, in taking the difference, the right-hand side becomes divisible by $\alpha^2 - \beta^2$, and therefore in the final result we have on the left-hand side the simple factor $\alpha^2 - \beta^2$ instead of $(\alpha^2 - \beta^2)^2$.

Similarly

$$\begin{aligned}(\alpha^2 - \beta^2)^2 YX' &= \alpha\beta (C^2 u C^2 u' + D^2 u D^2 u') - \alpha^2 D^2 u C^2 u' - \beta^2 C^2 u D^2 u', \\ XY' &= \alpha\beta \text{ " } - \beta^2 \text{ " } - \alpha^2 \text{ " },\end{aligned}$$

and thence

$$\begin{aligned}(\alpha^2 - \beta^2)^2(YX' + XY') &= 2\alpha\beta(C^2u C^2u' + D^2u D^2u') - (\alpha^2 + \beta^2)(C^2u D^2u' + D^2u C^2u'), \\(\alpha^2 - \beta^2)(-YX' + XY') &= D^2u C^2u' - C^2u D^2u'.\end{aligned}$$

48. Hence recollecting that

$$A^20 = \alpha^2 + \beta^2, \quad B^20 = 2\alpha\beta,$$

the original equations become

$$\begin{aligned}C^20 \cdot A(u + u')A(u - u') &= A^20(C^2u C^2u' + D^2u D^2u') - B^20(C^2u D^2u' + D^2u C^2u'), \\C^20 \cdot B(u + u')B(u - u') &= B^20(C^2u C^2u' + D^2u D^2u') - A^20(C^2u D^2u' + D^2u C^2u'), \\C^20 \cdot C(u + u')C(u - u') &= C^2u C^2u' - D^2u D^2u', \\C^20 \cdot D(u + u')D(u - u') &= D^2u C^2u' - C^2u D^2u'.\end{aligned}$$

49. It will be observed that the four products $A(u + u')A(u - u')$, &c., are each of them expressed in terms of C^2u, D^2u, C^2u', D^2u' . Since each of the squared functions A^2u, B^2u, C^2u, D^2u is a linear function of any two of them, and the like as regards $A^2u', B^2u', C^2u', D^2u'$, it is clear that in each equation we can on the right-hand side introduce any two at pleasure of the squared functions of u , and any two at pleasure of the squared functions of u' . But all the forms so obtained are of course identical if, taking x' the same function of u' that x is of u , we introduce on the right-hand side x, x' instead of u, u' ; and the values of $A(u + u')A(u - u')$, &c., are found to be equal to multiples of $\nabla, \nabla_1, \nabla_2, \nabla_3$, where

$$\nabla = x - x', \quad \nabla_1 = \begin{vmatrix} 1 & x + x' & xx' \\ 1 & a + d & ad \\ 1 & b + c & bc \end{vmatrix}, \quad \nabla_2 = \begin{vmatrix} 1 & x + x' & xx' \\ 1 & b + d & bd \\ 1 & c + a & ca \end{vmatrix}, \quad \nabla_3 = \begin{vmatrix} 1 & x + x' & xx' \\ 1 & c + d & cd \\ 1 & a + b & ab \end{vmatrix}.$$

50. In fact, from the equations

$$A^2u = \mathfrak{A}(a - x), \quad A^2u' = \mathfrak{A}(a - x'),$$

we have

$$\begin{aligned}\nabla &= \frac{1}{\mathfrak{A}\mathfrak{B}}(B^2u C^2u' - C^2u B^2u') = \frac{1}{\mathfrak{C}\mathfrak{A}}(C^2u A^2u' - A^2u C^2u') = \frac{1}{\mathfrak{A}\mathfrak{B}}(A^2u B^2u' - B^2u A^2u'), \\&= \frac{1}{\mathfrak{A}\mathfrak{D}}(A^2u D^2u' - D^2u A^2u') = -\frac{1}{\mathfrak{B}\mathfrak{D}}(B^2u D^2u' - D^2u B^2u') = \frac{1}{\mathfrak{C}\mathfrak{D}}(C^2u D^2u' - D^2u C^2u').\end{aligned}$$

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where it will be recollected that

$$f \mathfrak{A}\mathfrak{D} = a \mathfrak{B}\mathfrak{C}, \quad -g \mathfrak{B}\mathfrak{D} = b \mathfrak{C}\mathfrak{A}, \quad -h \mathfrak{C}\mathfrak{D} = c \mathfrak{A}\mathfrak{B}.$$

Moreover

$$\begin{aligned}(b - c)\nabla_1 &= \begin{vmatrix} b - x \cdot b - x' & c - x \cdot c - x' \\ b - a \cdot b - d & c - a \cdot c - d \end{vmatrix}, \quad (a - d)\nabla_1 = \begin{vmatrix} a - x \cdot a - x' & d - x \cdot d - x' \\ a - b \cdot a - c & d - b \cdot d - c \end{vmatrix}, \\(c - a)\nabla_2 &= -\begin{vmatrix} c - x \cdot c - x' & a - x \cdot a - x' \\ c - b \cdot c - d & a - b \cdot a - d \end{vmatrix}, \quad (b - d)\nabla_2 = \begin{vmatrix} b - x \cdot b - x' & d - x \cdot d - x' \\ b - c \cdot b - a & d - c \cdot d - a \end{vmatrix}, \\(a - b)\nabla_3 &= -\begin{vmatrix} a - x \cdot a - x' & b - x \cdot b - x' \\ a - c \cdot a - d & b - c \cdot b - d \end{vmatrix}, \quad (c - d)\nabla_3 = \begin{vmatrix} c - x \cdot c - x' & d - x \cdot d - x' \\ c - a \cdot c - b & d - a \cdot d - b \end{vmatrix},\end{aligned}$$

or as these may be written

$$\begin{aligned}\nabla_1 &= \frac{1}{f}\{bh.b - x.b - x' + cg.c - x.c - x'\} = \frac{1}{f}\{gh.a - x.a - x' + bc.d - x.d - x'\}, \\ \nabla_2 &= -\frac{1}{g}\{cf.c - x.c - x' + ah.a - x.a - x'\} = -\frac{1}{g}\{hf.b - x.b - x' + ca.d - x.d - x'\}, \\ \nabla_3 &= -\frac{1}{h}\{ag.a - x.a - x' + bf.b - x.b - x'\} = \frac{1}{h}\{fg.c - x.c - x' + ab.d - x.d - x'\},\end{aligned}$$

that is,

$$\begin{aligned}\nabla_1 &= -\frac{1}{a}\frac{1}{\mathfrak{B}^2}(\mathfrak{B}^2uB^2u' + C^2uC^2u') = -\frac{1}{f}\frac{1}{\mathfrak{A}\mathfrak{D}}(A^2uA^2u' + D^2uD^2u'), \\ \nabla_2 &= -\frac{1}{b}\frac{1}{\mathfrak{C}^2}(C^2uC^2u' - A^2uA^2u') = \frac{1}{g}\frac{1}{\mathfrak{B}\mathfrak{D}}(B^2uD^2u' - D^2uD^2u'), \\ \nabla_3 &= -\frac{1}{af}(-A^2uA^2u' + B^2uB^2u') = \frac{1}{h}\frac{1}{\mathfrak{C}\mathfrak{D}}(C^2uC^2u' - B^2uB^2u').\end{aligned}$$

51. Hence $\nabla, \nabla_1, \nabla_2, \nabla_3$ denoting these functions of x, x' or of u, u' , we have

$$\begin{aligned}A(u + u')A(u - u') &= \frac{\mathfrak{A}}{gh}\nabla_1, \\ B(u + u')B(u - u') &= \frac{\mathfrak{B}}{hf}\nabla_2, \\ C(u + u')C(u - u') &= -\frac{\mathfrak{C}}{fg}\nabla_3, \\ D(u + u')D(u - u') &= \mathfrak{D}\nabla.\end{aligned}$$

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The square-set $u \pm u'$, u' indefinitely small: differential formulæ of the second order.

52. I consider the original form

$$C^20C(u + u')C(u - u') = C^2uC^2u' - D^2uD^2u',$$

which is of course included in the last-mentioned equations.

Writing this in the form

$$\frac{C^20C(u + u')C(u - u')}{C^2u} = C^2u' - \frac{D^2uD^2u'}{C^2u},$$

and taking u' indefinitely small, whence

$$\begin{aligned}C(u + u') &= Cu + u'C'u + \frac{1}{2}u'^2C''u, & Cu' &= C0, \\ C(u - u') &= Cu - u'C'u + \frac{1}{2}u'^2C''u, & Du' &= u'D'0, \\ C(u + u')C(u - u') &= C^2u + u'^2\{CuC''u - (C'u)^2\},\end{aligned}$$

the equation becomes

$$\frac{C^20\{C^2u + u'^2(CuC''u - (C'u)^2)\}}{C^2u} = C^20 - u'^2\frac{D^2uD^20}{C^2u},$$

that is,

$$\frac{C''u}{Cu} - \left(\frac{C'u}{Cu}\right)^2 = \frac{C''0}{C0} - \left(\frac{D'0}{C0}\right)^2\frac{D^2u}{C^2u},$$

viz. we have $\left(\frac{d}{du}\right)^2 \log Cu$ expressed in terms of the quotient-function $\frac{D^2u}{C^2u}$, and consequently Cu given as an exponential, the argument of which depends on the double integral $\int du \int du \frac{D^2u}{C^2u}$.

53. To complete the result, I write the equation in the form

$$\frac{d^2}{du^2} \log Cu = \frac{C''0}{C0} - \frac{1}{k}\left(\frac{B'0}{C0}\right)^2 + \frac{1}{k}\left(\frac{D'0}{C0}\right)^2\left(1 - k\frac{D^2u}{C^2u}\right);$$

$\frac{D'0}{C0}$ is $= -\sqrt{k}K$, and $\frac{C''0}{C0}$ is $= K(K-E)$; hence the equation is

$$\frac{d^2}{du^2} \log Cu = K^2 \left(1 - \frac{E}{K}\right) - k^2 \left(\frac{D^2u}{C^2u}\right), = K^2 \left(1 - \frac{E}{K} - k^2 \operatorname{sn}^2 Ku\right),$$

or integrating twice, and observing that $\frac{d}{du} \log Cu$ and $\log Cu$, for $u = 0$, become $= 0$ and $\log C0$ respectively, we have

$$\log Cu = \log C0 + \frac{1}{2} \left(1 - \frac{E}{K}\right) K^2 u^2 - k^2 \int_0^u du \int_0^u du K^2 \operatorname{sn}^2 Ku,$$

which is in fact

$$\log \Theta(Ku) = \log C0 + \frac{1}{2} \left(1 - \frac{E}{K}\right) K^2 u^2 - k^2 \int_0^u du \int_0^u du K^2 \operatorname{sn}^2 Ku,$$

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agreeing with Jacobi's formula

$$\log \Theta u = \log \Theta 0 + \frac{1}{2} \left(1 - \frac{E}{K}\right) u^2 - k^2 \int_0^u du \int_0^u du \operatorname{sn}^2 u.$$

Elliptic integrals of the third kind.

54. We may write

$$\begin{aligned} \frac{A(u+u')A(u-u')}{A^2u A^2u'} &= \frac{1}{\mathfrak{A}gh} \frac{\nabla_1}{a-x \cdot a-x'}, \\ \frac{B(u+u')B(u-u')}{B^2u B^2u'} &= \frac{1}{\mathfrak{B}hf} \frac{\nabla_2}{b-x \cdot b-x'}, \\ \frac{C(u+u')C(u-u')}{C^2u C^2u'} &= \frac{1}{\mathfrak{C}fg} \frac{\nabla_3}{c-x \cdot c-x'}, \\ \frac{D(u+u')D(u-u')}{D^2u D^2u'} &= \frac{1}{\mathfrak{D}} \frac{x-x'}{d-x \cdot d-x'}. \end{aligned}$$

We differentiate logarithmically in regard to u' . Observing that

$$K du' = \frac{1}{2} \frac{\sqrt{-af} dx'}{\sqrt{a-x' \cdot b-x' \cdot c-x' \cdot d-x'}} = \frac{1}{2} \frac{\sqrt{-af}}{\sqrt{X'}} dx',$$

suppose, the first equation gives

$$-\frac{A'u'}{Au'} + \frac{A'(u-u')}{A(u-u')} - \frac{A'(u+u')}{A(u+u')} = \frac{2K\sqrt{X'}}{\sqrt{-af}} \frac{d}{dx'} \log \frac{\nabla_1}{a-x'};$$

and if for a moment

$$\nabla_1 = \begin{vmatrix} 1 & x+x' & xx' \\ 1 & a+d & ad \\ 1 & b+c & bc \end{vmatrix}$$

is put

$$= P(a-x') + Q(d-x'),$$

then

$$\frac{d}{dx'} \log \frac{\nabla_1}{a-x'} = \frac{d}{dx'} \log \nabla_1 + \frac{1}{a-x'} = \frac{P+Q \frac{d-x'}{a-x'}}{(a-x')\nabla_1} - \dots = -\frac{Q(d-x')}{(a-x')\nabla_1}.$$

But, writing $x' = a$, we have

$$Q(d-a) = -Qf = \begin{vmatrix} 1 & a+x & ax \\ 1 & a+d & ad \end{vmatrix} = -(a-b)(a-c)(d-x) = -bc(d-x),$$

that is,

$$Qf = -bc(d - x),$$

or

$$\frac{d}{dx'} \log \frac{\nabla_1}{a - x'} = \frac{bc(d - x)}{(a - x')\nabla_1}.$$

Hence the equation is

$$\frac{A'(u')}{A(u')} - \frac{A'(u - u')}{A(u - u')} - \frac{A'(u + u')}{A(u + u')} = \frac{2K\sqrt{X'}}{\sqrt{-af}} \cdot \frac{bc(d - x)}{(a - x')\nabla_1}.$$

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and similarly

$$\begin{aligned} \frac{B'(u')}{B(u')} + \frac{B'(u - u')}{B(u - u')} - \frac{B'(u + u')}{B(u + u')} &= \frac{2K ca}{\sqrt{-af}} \sqrt{X'} \cdot \frac{d - x}{(b - x')\nabla_2}, \\ \frac{C'(u')}{C(u')} + \frac{C'(u - u')}{C(u - u')} - \frac{C'(u + u')}{C(u + u')} &= \frac{2K ab}{\sqrt{-af}} \sqrt{X'} \cdot \frac{d - x}{(c - x')\nabla_3}, \\ \frac{D'(u')}{D(u')} + \frac{D'(u - u')}{D(u - u')} - \frac{D'(u + u')}{D(u + u')} &= \frac{2K}{\sqrt{-af}} \sqrt{X'} \cdot \frac{d - x}{(d - x')(x - x')}. \end{aligned}$$

55. Multiply each of these equations by $du, = \frac{1}{2} \frac{\sqrt{-af}}{K} \frac{dx}{\sqrt{X}}$, and integrate. We have equations such as

$$2u \frac{A'(u')}{A(u')} + \log \frac{A(u - u')}{A(u + u')} = \text{const.} + \frac{1}{\sqrt{-af}} \int \frac{bc(d - x) dx}{(a - x')\nabla_1 \sqrt{X}},$$

showing how the integrals of the third kind

$$\int \frac{(d - x) dx}{\nabla_1 \sqrt{X}}, \quad \int \frac{(d - x) dx}{\nabla_2 \sqrt{X}}, \quad \int \frac{(d - x) dx}{\nabla_3 \sqrt{X}}, \quad \int \frac{(d - x) dx}{(x - x')\sqrt{X}}$$

depend on the theta-functions.

If, instead, we work with the original equation

$$\frac{C(u + u')C(u - u')}{Cu \cdot C^2u'} = 1 - \frac{D^2u}{C^2u} \frac{D^2u'}{C^2u'},$$

we find in the same way

$$\begin{aligned} \frac{C'(u')}{C(u')} + \frac{C'(u - u')}{C(u - u')} - \frac{C'(u + u')}{C(u + u')} &= -\frac{d}{du'} \log \left(1 - \frac{D^2u}{C^2u} \frac{D^2u'}{C^2u'} \right), \\ &= -\frac{d}{du'} \log(1 - k^2 \text{sn}^2 Ku \text{sn}^2 Ku'), \\ &= -\frac{2k^2 K \text{sn} Ku' \text{cn} Ku' \text{dn} Ku' \text{sn}^2 Ku}{1 - k^2 \text{sn}^2 Ku' \text{sn}^2 Ku}, \end{aligned}$$

or, multiplying by $\frac{1}{2} du$ and integrating, we have

$$\frac{C'(u')}{C(u')} + \frac{1}{2} \log \frac{C(u - u')}{C(u + u')} = - \int \frac{k^2 \text{sn} Ku' \text{cn} Ku' \text{dn} Ku' \text{sn}^2 Ku}{1 - k^2 \text{sn}^2 Ku' \text{sn}^2 Ku} K du,$$

which is in fact Jacobi's equation

$$\frac{\Theta'a}{\Theta a} + \frac{1}{2} \log \frac{\Theta(u-a)}{\Theta(u+a)} = k^2 \int \frac{\operatorname{sn} a \operatorname{cn} a \operatorname{dn} a \operatorname{sn}^2 u \, du}{1 - k^2 \operatorname{sn}^2 a \operatorname{sn}^2 u}, \quad = \Pi(u, a).$$

I do not effect the operation but consider the forms first obtained,

$$A(u+u')A(u-u') = \frac{2}{\operatorname{gh}} \nabla_1, \quad \&c.,$$

as the equivalent of Jacobi's last-mentioned equation.

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Addition-formulæ.

56. The addition-theorem for the quotient-functions is of course given by means of the theorem for the elliptic functions: but more elegantly by the formulæ relating to the form $dx \div \sqrt{a-x} \cdot \sqrt{b-x} \cdot \sqrt{c-x} \cdot \sqrt{d-x}$ obtained in my paper "On the Double ϑ -Functions" (Crelle, t. LXXXVII. (1879), pp. 74–81, [697]); viz. for the differential equation

$$\frac{dx}{\sqrt{X}} + \frac{dy}{\sqrt{Y}} + \frac{dz}{\sqrt{Z}} = 0,$$

to obtain the particular integral which for $y = d$ reduces itself to $z = x$, we must, in the formulæ of the paper just referred to, interchange a and d : and writing for shortness $a, b, c, d = a-x, b-x, c-x, d-x$, and similarly $a_1, b_1, c_1, d_1 = a-y, b-y, c-y, d-y$, then when the interchange is made, the formulæ become

$$\begin{aligned} \sqrt{\frac{d-z}{a-z}} &= \frac{\sqrt{\frac{d-b}{d-a}} \{ (d-c) \sqrt{abc_1 d_1} + (b-a) \sqrt{cda_1 b_1} \}}{(bc, ad)} = \frac{\sqrt{\frac{d-b}{d-a}} (ac, bd)}{\sqrt{abc_1 d_1} - \sqrt{cda_1 b_1}}, \\ \sqrt{\frac{b-z}{a-z}} &= \frac{\sqrt{\frac{d-c}{d-a}} (ac, bd)}{\sqrt{bda_1 c_1} - \sqrt{acb_1 d_1}} = \frac{\sqrt{\frac{d-b}{d-c}} \{ (d-c) \sqrt{aba_1 b_1} - (b-a) \sqrt{cdc_1 d_1} \}}{(d-c) \sqrt{aba_1 b_1} + (b-a) \sqrt{cdc_1 d_1}} \sqrt{\frac{d-b}{d-c}}, \\ \sqrt{\frac{c-z}{a-z}} &= \frac{\sqrt{\frac{d-b}{d-c}} \{ (d-a) \sqrt{bca_1 c_1} - (b-c) \sqrt{ada_1 d_1} \}}{(d-b) \sqrt{aca_1 c_1} - (c-a) \sqrt{bdb_1 d_1}} = \frac{\sqrt{\frac{d-b}{d-c}} (ab, cd)}{(d-b) \sqrt{aca_1 c_1} - (c-a) \sqrt{bdb_1 d_1}}, \\ \sqrt{\frac{d-z}{a-z}} \cdot \frac{1}{x-y} \cdot \frac{1}{(bc, ad)} &= \frac{\sqrt{abc_1 d_1} - \sqrt{cda_1 b_1}}{\sqrt{abc_1 d_1} - \sqrt{cda_1 b_1}}. \end{aligned}$$

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57. In the foregoing formulæ, (bc, ad) , (ac, bd) , and (ab, cd) denote respectively

$$\begin{vmatrix} 1 & x+y & xy \\ 1 & b+c & bc \\ 1 & a+d & ad \end{vmatrix}, \quad \begin{vmatrix} 1 & x+y & xy \\ 1 & c+a & ca \\ 1 & b+d & bd \end{vmatrix}, \quad \begin{vmatrix} 1 & x+y & xy \\ 1 & a+b & ab \\ 1 & c+d & cd \end{vmatrix};$$

and substituting for $\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{D}$ their values, and for a, b , &c., writing again $a - x$, $b - x$, &c., we have moreover

$$\begin{aligned} A^2u &= \sqrt{c-b \cdot b-d \cdot c-d}(a-x), & A^2v &= \cdots (a-y), \\ B^2u &= \sqrt{c-a \cdot c-d \cdot a-d}(b-x), & B^2v &= \cdots (b-y), \\ C^2u &= \sqrt{a-b \cdot a-d \cdot b-d}(c-x), & C^2v &= \cdots (c-y), \\ D^2u &= \sqrt{c-b \cdot c-a \cdot a-b}(d-x), & D^2v &= \cdots (d-y), \\ A^2(u+v) &= \cdots (a-z), \\ B^2(u+v) &= \cdots (b-z), \\ C^2(u+v) &= \cdots (c-z), \\ D^2(u+v) &= \cdots (d-z), \end{aligned}$$

the constant multipliers being of course the same in the three columns respectively. According to what precedes, the radical of the fourth line should be taken with the sign $-$. The functions (bc, ad) , &c., contained in the formulæ, require a transformation such as

$$(b-c)(bc, ad) = \begin{vmatrix} b-x \cdot b-y, & c-x \cdot c-y \\ b-a \cdot b-d, & c-a \cdot c-d \end{vmatrix}$$

in order to make them separately homogeneous in the differences $a-x, b-x, c-x, d-x$, and $a-y, b-y, c-y, d-y$, and therefore to make them expressible as linear functions of the squared functions A^2u , &c., and A^2v , &c., respectively: the formulæ then give the quotient-functions $\frac{A(u+v)}{D(u+v)}$, &c., in terms of the quotient-functions of u and v respectively.

Doubly infinite product-forms.

58. The functions Au, Bu, Cu, Du may be expressed each as a doubly infinite product. Writing for shortness

$$\begin{aligned} m+n \cdot \frac{\omega}{\pi i} &= (m, n), \\ m+1+n \cdot \frac{\omega}{\pi i} &= (\overline{m}, n), \\ m+(n+1) \cdot \frac{\omega}{\pi i} &= (m, \overline{n}), \\ m+1+(n+1) \cdot \frac{\omega}{\pi i} &= (\overline{m}, \overline{n}), \end{aligned}$$

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then the formulæ are

$$\begin{aligned} Au &= A0 \cdot \prod \prod \left\{ 1 + \frac{u}{(\overline{m}, \overline{n})} \right\}, \\ Bu &= B0 \cdot \prod \prod \left\{ 1 + \frac{u}{(m, \overline{n})} \right\}, \\ Cu &= C0 \cdot \prod \prod \left\{ 1 + \frac{u}{(\overline{m}, n)} \right\}, \\ Du &= D'0 \cdot u \cdot \prod \prod \left\{ 1 + \frac{u}{(m, n)} \right\}, \end{aligned}$$

where in all the formulæ, m and n denote even integers having all values whatever, zero included, from $-\infty$ to $+\infty$; only in the formula for Du , the term for which m and n are simultaneously $= 0$, is to be omitted.

59. But a further definition in regard to the limits is required: first, we assume that m has the corresponding positive and negative values, and similarly that n has corresponding positive and negative

values¹; or say, in the four formulæ respectively, we consider m, n as extending

$$\begin{array}{ll} m \text{ from } -\mu \text{ to } \mu + 2, & n \text{ from } -\nu \text{ to } \nu + 2, \\ -\mu \text{ to } \mu + 2, & -\nu \text{ to } \nu, \\ -\mu \text{ to } \mu, & -\nu \text{ to } \nu + 2, \\ -\mu \text{ to } \mu, & -\nu \text{ to } \nu, \end{array}$$

where μ and ν are each of them ultimately infinite. But, secondly, it is necessary that μ should be indefinitely larger than ν , or say that ultimately $\frac{\nu}{\mu} = 0$.

60. In fact, transforming the q -series into products as in the *Fundamenta Nova*, and neglecting for the moment mere constant factors, we have

$$\begin{aligned} Au &= (1 + 2q \cos \pi u + q^2)(1 + 2q^3 \cos \pi u + q^6) \cdots, \\ Bu &= \cos \frac{1}{2}\pi u (1 + 2q^2 \cos \pi u + q^4)(1 + 2q^4 \cos \pi u + q^8) \cdots, \\ Cu &= (1 - 2q \cos \pi u + q^2)(1 - 2q^3 \cos \pi u + q^6) \cdots, \\ Du &= \sin \frac{1}{2}\pi u (1 - 2q^2 \cos \pi u + q^4)(1 - 2q^4 \cos \pi u + q^8) \cdots, \end{aligned}$$

and writing for a moment $\omega = -\frac{\omega}{\pi i}$ and therefore $q^{1+n} + q^{-1-n} = e^{i\pi\alpha} + e^{-i\pi\alpha} = 2 \cos \frac{1}{2}\pi\alpha$, &c., each of these expressions is readily converted into a singly infinite product of sines or cosines

$$\begin{aligned} Au &= \prod \cos \frac{1}{2}\pi(u + n\alpha), \\ Bu &= \prod \cos \frac{1}{2}\pi(u + \bar{n}\alpha), \\ Cu &= \prod \sin \frac{1}{2}\pi(u + n\alpha), \\ Du &= \prod \sin \frac{1}{2}\pi(u + \bar{n}\alpha), \end{aligned}$$

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where $\bar{n}$ is written to denote $n + 1$, and n has all positive or negative even values (zero included) from $-\infty$ to $+\infty$, or more accurately from $-\nu$ to ν , if ν is ultimately infinite.

61. The sines and cosines are converted into infinite products by the ordinary formulæ, which neglecting constant factors may conveniently be written

$$\sin \frac{1}{2}\pi u = \prod (u + m), \quad \cos \frac{1}{2}\pi u = \prod (u + \bar{m}),$$

where $\bar{m}$ is written to denote $m + 1$, and m has all positive or negative even values (zero included) from $-\infty$ to $+\infty$, or more accurately from $-\mu$ to μ , if μ be ultimately infinite. But in applying these formulæ to the case of a function such as

$$\sin \frac{1}{2}\pi(u + n\alpha),$$

it is assumed that the limiting values $\mu, -\mu$ of m are indefinitely large in regard to $u + n\alpha$; and therefore, since n may approach to its limiting value $+\nu$, it is necessary that μ should be indefinitely large in comparison with ν , or that $\frac{\nu}{\mu} = 0$.

62. It is on account of this unsymmetry as to the limits $\frac{\nu}{\mu} = 0, \frac{\mu}{\nu} = \infty$, that we have 1 as a quarter-period, $\frac{\omega}{\pi i}$ only as a quarter-quasi-period of the single theta-functions.

$$\text{The transformation } q \text{ to } r, \log q \log r = \pi^2.$$

¹This is more than is necessary; it would be enough if the ultimate values of m were $-\mu, \mu'$, and μ, μ' being in a ratio of equality; and the like as regards n . But it is convenient that the numbers should be absolutely equal.

63. In general, if we consider the ratio of two such infinite products, where for the first the limits are $(\pm\mu, \pm\nu)$, and for the second they are $(\pm\mu', \pm\nu')$, and where for convenience we take $\mu > \mu', \nu > \nu'$, then the quotient, say $[\mu, \nu] \div [\mu', \nu']$ is $= \exp(Mu^2)$, where

$$M = -\frac{1}{2} \iint \frac{dm \, dn}{(m, n)^2},$$

taken over the area included between the two rectangles. We have

$$(m, n) = m + \frac{\omega}{\pi i} n = m + i\theta n$$

suppose, where (ω being negative) $\theta = -\frac{\omega}{\pi}$ is positive: the integral is

$$\iint \frac{dm \, dn}{(m + i\theta n)^2} = \int dm \int \frac{dn}{(m + i\theta n)^2} = -\frac{1}{i\theta} \int dm \left(\frac{1}{m - i\theta\nu} - \frac{1}{m + i\theta\nu} \right) = -\frac{1}{i\theta} \int \log \frac{m - i\theta\nu}{m + i\theta\nu} dm,$$

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or finally between the proper limits the value is

$$= \frac{2}{i\theta} \left\{ \log \left(\frac{\mu - i\theta\nu}{\mu + i\theta\nu} \right) - \log \left(\frac{\mu' - i\theta\nu'}{\mu' + i\theta\nu'} \right) \right\},$$

where the logarithms are

$$\log(\mu - i\theta\nu) = \log \sqrt{\mu^2 + \nu^2} - i \tan^{-1} \frac{\theta\nu}{\mu} - \&c.,$$

and the $\tan^{-1}$ denotes always an arc between the limits $-\frac{1}{2}\pi, +\frac{1}{2}\pi$. Hence, if $\frac{\mu}{\nu} = \infty, \frac{\mu'}{\nu'} = 0$, the value is

$$\frac{2}{i\theta} (-0 - 0 + \frac{1}{2}\pi i + \frac{1}{2}\pi i) = \frac{2\pi}{\theta} = -\frac{2\pi^2}{\omega} = \text{or } \frac{2\pi^2}{a};$$

Hence finally

$$[\mu, \nu, = \infty] \div [\mu, \nu, = 0] = \exp\left(\frac{\pi^2}{a} u^2\right).$$

64. We have $a = \log q$, negative; hence taking r such that $\log q \log r = \pi^2$, we have $a' = \log r$, also negative; and r , like q , is positive and less than 1. We consider the theta-functions which depend on r in the same manner that the original functions did on q , say these are

$$\begin{aligned} A(u, r) &= A(0, r) \prod \prod \left\{ 1 + \frac{u}{m + n \frac{a'}{\pi i}} \right\}, \\ B(u, r) &= B(0, r) \prod \prod \left\{ 1 + \frac{u}{\bar{m} + n \frac{a'}{\pi i}} \right\}, \\ C(u, r) &= C(0, r) \prod \prod \left\{ 1 + \frac{u}{\bar{m} + \bar{n} \frac{a'}{\pi i}} \right\}, \\ D(u, r) &= D'(0, r) u \prod \prod \left\{ 1 + \frac{u}{m + \bar{n} \frac{a'}{\pi i}} \right\}, \end{aligned}$$

limits as before, and in particular $\frac{\mu}{\nu} = \infty$; it is at once seen that if in the original functions, which I now call $A(u, q), B(u, q), C(u, q), D(u, q)$, we write $\frac{au}{\pi i}$ for u , we obtain the same infinite products which present themselves in the expressions of the new functions $A(u, r)$, &c., only with a different condition as to the limits; we have for instance

$$\prod \prod \left\{ 1 + \frac{\frac{au}{\pi i}}{m + n \frac{a'}{\pi i}} \right\} = \prod \prod \left\{ 1 + \frac{u}{n + m \frac{\pi i}{a}} \right\} = \prod \prod \left\{ 1 + \frac{u}{n + m \frac{\pi i}{a'}} \right\}$$

which, interchanging m, n , and of course also μ, ν , is

$$= \prod \prod \left\{ 1 + \frac{u}{m + n \frac{a'}{\pi i}} \right\}$$

with the condition $\frac{\mu}{\nu} = 0$ instead of $\frac{\mu}{\nu} = \infty$. Hence disregarding for the moment constant factors, and observing that a is replaced by a' , we have

$$\begin{aligned} D(u, r) \div D\left(\frac{au}{\pi i}, q\right) &= [\mu, \nu, = \infty] \div [\mu, \nu, = 0] \\ &= \exp\left(\frac{\pi^2}{a} u^2\right) = \exp\left(\frac{1}{2} u^2 \log q\right). \end{aligned}$$

65. We have equations of this form for the four functions, but with a proper constant multiplier in each equation: the equations, in fact, are

$$\begin{aligned} A(u, r) &= \{A(0, r) \div A(0, q)\} \exp\left(\frac{1}{2} u^2 \log q\right) A\left(\frac{au}{\pi i}, q\right), \\ B(u, r) &= \{B(0, r) \div B(0, q)\} B\left(\frac{au}{\pi i}, q\right), \\ C(u, r) &= \{C(0, r) \div C(0, q)\} C\left(\frac{au}{\pi i}, q\right), \\ D(u, r) &= \{D'(0, r) \div D'(0, q)\} D\left(\frac{au}{\pi i}, q\right). \end{aligned}$$

It is to be observed that r is the same function of k' that q is of k . This would at once follow from Jacobi's equation $\log q = -\frac{\pi K'}{K}$, for then $\log q \log r = \pi^2$ and therefore $\log r = -\frac{\pi K}{K'}$ (only we are not at liberty to use the relation in question $\log q = -\frac{\pi K'}{K}$): assuming it to be true, we have

$$\begin{aligned} \frac{D'(0, r)}{A^2(0, r)} &= \frac{D'(0, q)}{C^2(0, q)} = \frac{C^2(0, q) D'(0, q)}{A(0, q) B(0, q)}, \\ &= \frac{C'(0, r)}{A^2(0, r)} = \frac{B'(0, r)}{B^2(0, r)} = \frac{A(0, r) D'(0, r)}{B(0, r) C(0, r)}, \\ \log q &= -\frac{\pi K'}{K}, \quad \log r = -\frac{\pi K}{K'}, \end{aligned}$$

where, if the identity of the two values of k or of the two values of k' were proved independently (as might doubtless be done), the required theorem, viz. that r is the same function of y that q is of k , would follow conversely: and thence the other equations of the system.

66. We have, in the *Fundamenta Nova*, p. 175, [Jacobi's *Ges. Werke*, t. i., p. 227], the equation

$$\frac{H(iu, k)}{\Theta(0, k)} = i \sqrt{\frac{k}{k'}} e^{\frac{\pi u^2}{4K}} \frac{H(u, k')}{\Theta(0, k')},$$

writing here $K'u$ instead of u the equation becomes

$$\frac{H(iK'u, k)}{\Theta(0, k)} = i \sqrt{\frac{k}{k'}} \exp\left(\frac{1}{4} \frac{\pi K'}{K} u^2\right) \cdot \frac{H(K'u, k')}{\Theta(0, k')},$$

or, what is the same thing,

$$\frac{D\left(\frac{\pi i K'}{K} u, q\right)}{C^2(0, q)} = i \sqrt{\frac{k}{k'}} \exp\left(\frac{1}{4} \frac{\pi K'}{K} u^2\right) \cdot \frac{D(u, r)}{D(0, r)},$$

which can be readily identified with the foregoing equation between $D\left(\frac{au}{\pi i}, q\right)$ and $D(u, r)$. But the real meaning of the transformation is best seen by means of the double-product formulæ.

THIRD PART.—THE DOUBLE THETA-FUNCTIONS.

Notations, &c.

67. We have here 16 functions $\vartheta\left(\begin{smallmatrix}\alpha\beta\\ \gamma\delta\end{smallmatrix}\right)(u,v)$: this notation by characteristics, containing each of them four numbers, is too cumbrous for ordinary use, and I therefore replace it by the current-number notation, in which the characteristics are denoted by the series of numbers 0, 1, 2, . . . , 15: we cannot in place of this introduce the single-and-double-letter notation $A, B, \dots, AB$, &c., for there is not here any correspondence of the two notations, nor is there anything in the definition of the functions which in anywise suggests the single-and-double-letter notation: this first presents itself in connexion with the relations between the functions given by the product-theorem: and as the product-theorem is based upon the notation by characteristics, it is proper to present the theorem in this notation, or in the equivalent current-number notation: and then to show how by the relations thus obtained between the functions we are led to the single-and-double-letter notation.

68. There are some other notations which may be referred to: and for showing the correspondence between them I annex the following table:

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The double theta-functions.

1.	2.	3.	4.	5.	6.	7.	8.
Asterisk denotes the odd functions.	Current number.	Single-and- double-letter, Cayley.	Göpel.	Göpel Cayley.	Rosenhain.	Weier- strass.	Kummer.
	0	$\begin{smallmatrix}00\\ 00\end{smallmatrix}$	BD	P'''	P_0	S_{11}	$\begin{smallmatrix}5\\ 12\end{smallmatrix}$
	1	$\begin{smallmatrix}10\\ 00\end{smallmatrix}$	CE	R'''	R_1	11	$\begin{smallmatrix}4\\ 8\end{smallmatrix}$
	2	$\begin{smallmatrix}01\\ 00\end{smallmatrix}$	CD	Q'''	Q_3	23	$\begin{smallmatrix}01\\ 10\end{smallmatrix}$
	3	$\begin{smallmatrix}11\\ 00\end{smallmatrix}$	BE	S'''	S_3	33	$\begin{smallmatrix}23\\ 6\end{smallmatrix}$
	4	$\begin{smallmatrix}00\\ 10\end{smallmatrix}$	AC	P''	A	02	$\begin{smallmatrix}34\\ 4\end{smallmatrix}$
*	5	$\begin{smallmatrix}10\\ 10\end{smallmatrix}$	C	iR''	R_2	12	$\begin{smallmatrix}3\\ 16\end{smallmatrix}$
	6	$\begin{smallmatrix}01\\ 10\end{smallmatrix}$	AB	Q''	Q_2	03	$\begin{smallmatrix}2\\ 2\end{smallmatrix}$
*	7	$\begin{smallmatrix}11\\ 10\end{smallmatrix}$	B	iS''	S_2	13	$\begin{smallmatrix}24\\ 14\end{smallmatrix}$
	8	$\begin{smallmatrix}00\\ 01\end{smallmatrix}$	BC	P'	A	20	$\begin{smallmatrix}12\\ 9\end{smallmatrix}$
	9	$\begin{smallmatrix}10\\ 01\end{smallmatrix}$	DE	R'	R_3	30	$\begin{smallmatrix}03\\ 5\end{smallmatrix}$
*	10	$\begin{smallmatrix}01\\ 01\end{smallmatrix}$	F	iQ'	Q_1	21	$\begin{smallmatrix}02\\ 11\end{smallmatrix}$
*	11	$\begin{smallmatrix}11\\ 01\end{smallmatrix}$	A	iS'	S_1	31	$\begin{smallmatrix}13\\ 7\end{smallmatrix}$
	12	$\begin{smallmatrix}00\\ 11\end{smallmatrix}$	AD	P	P	00	$\begin{smallmatrix}—\\ 1\end{smallmatrix}$
*	13	$\begin{smallmatrix}10\\ 11\end{smallmatrix}$	D	iR	R	10	$\begin{smallmatrix}04\\ 13\end{smallmatrix}$
	14	$\begin{smallmatrix}01\\ 11\end{smallmatrix}$	E	iQ	Q	01	$\begin{smallmatrix}1\\ 3\end{smallmatrix}$
	15	$\begin{smallmatrix}11\\ 11\end{smallmatrix}$	AE	S	S	11	$\begin{smallmatrix}14\\ 15\end{smallmatrix}$

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69. These are the notations:

1. By current-numbers. It may be remarked that the series was taken $0, 1, \dots, 15$, instead of $1, 2, \dots, 16$, in order that 0 might correspond to the characteristic $\begin{smallmatrix} 00 \\ 00 \end{smallmatrix}$;

2. By characteristics;

3. By single-and-double letters;

4. Göpel's, in his paper above referred to, and

5. The same as used in my paper above referred to;

6. Rosenhain's, in his paper above referred to;

7. Weierstrass', as quoted by Königsberger in his paper "Ueber die Transformation der Abelschen Functionen erster Ordnung," Crelle-Borchardt, t. LXIV. (1865), p. 17, and by Borchardt in his paper above referred to;

8. Not a theta-notation, but the series of current-numbers used in Kummer's Memoir "Ueber die algebraischen Strahlen-systeme," Berl. Abh. 1866, for the nodes of his 16-nodal quartic surface, and connected with the double theta-functions in my paper above referred to.

But in the present memoir only the first three columns of the table need be attended to.

70. It will be convenient to introduce here some other remarks as to notation, &c.

The letter c is used in connexion with the zero values $u = 0, v = 0$ of the arguments, viz.:

$$\vartheta_0, \vartheta_1, \vartheta_2, \vartheta_3, \vartheta_4, \vartheta_6, \vartheta_8, \vartheta_9, \vartheta_{12}, \vartheta_{15}$$

are even functions, and the corresponding zero-functions are denoted by

$$c_0, c_1, c_2, c_3, c_4, c_6, c_8, c_9, c_{12}, c_{15};$$

there are thus 10 constants c .

When (u, v) are indefinitely small each of these functions is of course equal to its zero-value plus a quadric term in (u, v) , and we may write in general

$$\vartheta = c + \frac{1}{2}(c^{[1]}, c^{[2]}, c^{[3]}|u, v)^2 :$$

there are thus 30 constants $c^{[1]}, c^{[2]}, c^{[3]}$.

The remaining functions

$$\vartheta_5, \vartheta_7, \vartheta_{10}, \vartheta_{11}, \vartheta_{13}, \vartheta_{14}$$

are odd functions vanishing for $u = 0, v = 0$; when these arguments are indefinitely small, we may write in general

$$\vartheta = (c', c''|u, v) :$$

there are thus 12 constants c', c'' .

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71. All these constants are in the first instance given as transcendental functions of the parameters, or say rather of $\exp a, \exp h, \exp b$, which exponentials correspond to the q of the single theory: viz., in a notation which will be readily understood, the constants $c, c^{[1]}, c^{[2]}, c^{[3]}$ of the even functions are

$$\Sigma \exp \left(\begin{matrix} m + \alpha, n + \beta \\ \gamma \quad \delta \end{matrix} \right),$$

$$-\frac{1}{4}\pi^2 \Sigma (m + \alpha)^2, 2(m + \alpha)(n + \beta), (n + \beta)^2, \exp \left(\begin{matrix} m + \alpha, n + \beta \\ \gamma \quad \delta \end{matrix} \right);$$

and the constants c', c'' of the odd functions are

$$i\pi \Sigma (m + \alpha), (n + \beta), \exp \left(\begin{matrix} \\ \delta \end{matrix} \right).$$

72. It is convenient for the verification of results and otherwise, to have the values of the functions, belonging to small values of $\exp(-a)$, $\exp(-b)$; if to avoid fractional exponents we regard these and $\exp(-h)$ as fourth powers, and write

$$\exp(-a) = Q^4, \quad \exp(-h) = R^4, \quad \exp(-b) = S^4,$$

also

$$QRS = \Lambda, \quad QR^{-1}S = \Lambda', \quad \text{and therefore } \Lambda\Lambda' = Q^2S^2,$$

then attending only to the lowest powers of Q, S we find without difficulty

$$\begin{aligned} \vartheta_0(u) &= 1, & \text{and therefore also } c_0 &= 1, \\ \vartheta_1 &= 2Q \cos \tfrac{1}{2}\pi u, & c_1 &= 2Q, \\ \vartheta_2 &= 2S \cos \tfrac{1}{2}\pi v, & c_2 &= 2S, \\ \vartheta_3 &= 2\Lambda \cos \tfrac{1}{2}\pi(u+v) + 2\Lambda' \cos \tfrac{1}{2}\pi(u-v), & c_3 &= 2\Lambda + 2\Lambda', \\ \vartheta_4 &= 1, & c_4 &= 1, \\ \vartheta_5 &= -2Q \sin \tfrac{1}{2}\pi u, & & \\ \vartheta_6 &= 2S \cos \tfrac{1}{2}\pi v, & c_6 &= 2S, \\ \vartheta_7 &= -2\Lambda \sin \tfrac{1}{2}\pi(u+v) - 2\Lambda' \sin \tfrac{1}{2}\pi(u-v), & & \\ \vartheta_8 &= 1, & c_8 &= 1, \\ \vartheta_9 &= 2Q \cos \tfrac{1}{2}\pi u, & c_9 &= 2Q, \\ \vartheta_{10} &= -2S \sin \tfrac{1}{2}\pi v, & & \\ \vartheta_{11} &= -2\Lambda \sin \tfrac{1}{2}\pi(u+v) + 2\Lambda' \sin \tfrac{1}{2}\pi(u-v), & & \\ \vartheta_{12} &= 1, & c_{12} &= 1, \\ \vartheta_{13} &= -2Q \sin \tfrac{1}{2}\pi u, & & \\ \vartheta_{14} &= -2S \sin \tfrac{1}{2}\pi v, & & \\ \vartheta_{15} &= -2\Lambda \cos \tfrac{1}{2}\pi(u+v) + 2\Lambda' \cos \tfrac{1}{2}\pi(u-v), & c_{15} &= -2\Lambda + 2\Lambda'. \end{aligned}$$

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73. The expansions might be carried further; we have for instance

$$\begin{aligned} \vartheta_0(u) &= 1 + 2Q^4 \cos \pi u + 2S^4 \cos \pi v, & c_0 &= 1 + 2Q^4 + 2S^4, \\ \vartheta_4 &= 1 - 2Q^4 & + 2S^4 &, & c_4 &= 1 - 2Q^4 + 2S^4, \\ \vartheta_8 &= 1 + 2Q^4 & - 2S^4 &, & c_8 &= 1 + 2Q^4 - 2S^4, \\ \vartheta_{12} &= 1 - 2Q^4 & - 2S^4 &, & c_{12} &= 1 - 2Q^4 - 2S^4, \end{aligned}$$

$$\begin{aligned} \vartheta_1 &= 2Q \cos \tfrac{1}{2}\pi u + 2Q^9 \cos \tfrac{3}{2}\pi u + 2A \cos \tfrac{1}{2}\pi(u+2v) + 2A' \cos \tfrac{1}{2}\pi(u-2v), & c_1 &= 2Q + 2Q^9 + 2A + 2A', \\ \vartheta_5 &= -2Q \sin \tfrac{1}{2}\pi u + 2Q^9 \sin \tfrac{3}{2}\pi u - 2A \sin \tfrac{1}{2}\pi(u+2v) - 2A' \sin \tfrac{1}{2}\pi(u-2v), \\ \vartheta_9 &= 2Q \cos \tfrac{1}{2}\pi u + 2Q^9 \cos \tfrac{3}{2}\pi u - 2A \cos \tfrac{1}{2}\pi(u+2v) - 2A' \cos \tfrac{1}{2}\pi(u-2v), & c_9 &= 2Q + 2Q^9 - 2A - 2A', \\ \vartheta_{13} &= -2Q \sin \tfrac{1}{2}\pi u + 2Q^9 \sin \tfrac{3}{2}\pi u + 2A \sin \tfrac{1}{2}\pi(u+2v) + 2A' \sin \tfrac{1}{2}\pi(u-2v), \end{aligned}$$

in which last formulæ

$$A = QR^4S^4 = \frac{Q}{R^3Q^3}, \quad A' = QR^{-4}S^4 = \frac{Q}{R^4S^4}.$$

74. In the single-and-double-letter notation we have six letters A, B, C, D, E, F ; and the duads $AB, AC, \dots, BE$ are used as abbreviations for the double triads ABF, CDE , &c., the letter F always accompanying the expressed duad; there are thus in all six single-letter symbols, and 10 double-letter symbols, in all 16 symbols, corresponding to the double-theta functions, as already mentioned in the order

$$\begin{array}{cccccccccccccccc} \vartheta_0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 & 11 & 12 & 13 & 14 & 15 \\ BD, & CE, & CD, & BE, & AC, & C, & AB, & B, & BC, & DE, & F, & A, & AD, & D, & E, & AE, \end{array}$$

where observe that the single letters C, B, F, A, D, E correspond to the odd functions 5, 7, 10, 11, 13, 14 respectively.

75. The ground of the notation is as follows:

The algebraical relations between the double theta-functions are such that, introducing six constant quantities a, b, c, d, e, f and two variable quantities x, y , it is allowable to express the 16 functions as proportional to given functions of these quantities $(a, b, c, d, e, f; x, y)$; viz. there are six functions the notations of which depend on the single letters a, b, c, d, e, f respectively, and 10 functions the notations of which depend on the pairs $ab, ac, ad, ae, bc, bd, be, cd, ce, de$ respectively: each of the 16 functions is, in fact, proportional to the product of two factors, viz. a constant factor depending only on (a, b, c, d, e, f) , and a variable factor containing also x and y . Attending in the first instance to the variable factors, and writing for shortness

$$\begin{aligned} a - x, b - x, c - x, d - x, e - x, f - x &= a, b, c, d, e, f; \quad x - y = \theta; \\ a - y, b - y, c - y, d - y, e - y, f - y &= a_1, b_1, c_1, d_1, e_1, f_1; \end{aligned}$$

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these are of the forms

$$\sqrt[4]{a} = \sqrt[4]{aa_1}, \quad \sqrt[4]{ab} = \frac{1}{\theta} \{ \sqrt{abfc_1d_1e_1} + \sqrt{a_1b_1f_1cde} \}.$$

I remark that to avoid ambiguity the squares of these expressions are throughout written as $(\sqrt[4]{a})^2$ and $(\sqrt[4]{ab})^2$ respectively.

76. There is, for the constant factors, a like single-and-double-letter notation which will be mentioned presently; but in the first instance I use for the even functions the before-mentioned 10 letters c , and for the odd ones six letters k . I assume that the values $x, y = \infty, \infty$ (ratio not determined) correspond to the values $u = 0, v = 0$ of the arguments; and that ω is a function of (x, y) which, when (x, y) are interchanged, changes only its sign, and which for indefinitely large values of (x, y) becomes $= \frac{x - y}{(xy)^{\frac{1}{2}}}$. This being so, we write (adding a second column which will be afterwards explained)

$$\begin{aligned} 0 &= BD = \omega c_0 \sqrt[4]{bd}, & c_0 &= \omega \sqrt[4]{bd}, \\ 1 &= CE = \omega c_1 \sqrt[4]{ce}, & c_1 &= \omega \sqrt[4]{ce}, \\ 2 &= CD = \omega c_2 \sqrt[4]{cd}, & c_2 &= \omega \sqrt[4]{cd}, \\ 3 &= BE = \omega c_3 \sqrt[4]{be}, & c_3 &= \omega \sqrt[4]{be}, \\ 4 &= AC = \omega c_4 \sqrt[4]{ac}, & c_4 &= \omega \sqrt[4]{ac}, \\ 5 &= C = \omega k_5 \sqrt[4]{c}, & k_5 &= \omega \sqrt[4]{c}, \\ 6 &= AB = \omega c_6 \sqrt[4]{ab}, & c_6 &= \omega \sqrt[4]{ab}, \\ 7 &= B = \omega k_7 \sqrt[4]{b}, & k_7 &= \omega \sqrt[4]{b}, \\ 8 &= BC = \omega c_8 \sqrt[4]{bc}, & c_8 &= \omega \sqrt[4]{bc}, \\ 9 &= DE = \omega c_9 \sqrt[4]{de}, & c_9 &= \omega \sqrt[4]{de}, \\ 10 &= F = \omega k_{10} \sqrt[4]{f}, & k_{10} &= \omega \sqrt[4]{f}, \\ 11 &= A = \omega k_{11} \sqrt[4]{a}, & k_{11} &= \omega \sqrt[4]{a}, \\ 12 &= AD = \omega c_{12} \sqrt[4]{ad}, & c_{12} &= \omega \sqrt[4]{ad}, \\ 13 &= D = \omega k_{13} \sqrt[4]{d}, & k_{13} &= \omega \sqrt[4]{d}, \\ 14 &= E = \omega k_{14} \sqrt[4]{e}, & k_{14} &= \omega \sqrt[4]{e}, \\ 15 &= AE = \omega c_{15} \sqrt[4]{ae}, & c_{15} &= \omega \sqrt[4]{ae}; \end{aligned}$$

viz. here, on writing $x, y = \infty, \infty$, each of the functions $\sqrt[4]{bd}$, &c., becomes $= \frac{(xy)^{\frac{1}{2}}}{x - y}$, and each of the functions $\sqrt[4]{a}$, &c., becomes $= \sqrt[4]{xy}$; hence by reason of the assumed

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form of ω , the odd functions each vanish (their evanescent values being proportional to $k_5, k_7, k_{10}, k_{11}, k_{13}, k_{14}$ respectively), while the even functions become equal to $c_0, c_1, c_2, c_3, c_4, c_6, c_8, c_9, c_{12}, c_{15}$ respectively.

Observe further that on interchanging x, y , the even functions remain unaltered, while the odd functions change their sign; that is, the interchange of x, y corresponds to the change u, v into $-u, -v$.

77. As to the values of the 10 c 's and the six k 's in terms of a, b, c, d, e, f , these are proportional to fourth roots, $\sqrt[4]{a}$, &c., $\sqrt[4]{ab}$, &c.; in $\sqrt[4]{a}$, a is in the first instance regarded as standing for the pentad $bcdef$, and then this is used to denote a product of differences $bc \cdot bd \cdot be \cdot bf \cdot cd \cdot ce \cdot cf \cdot de \cdot df \cdot ef$; similarly ab is in the first instance regarded as standing for the double triad $abf \cdot cde$, and then each of these triads is used to denote a product of differences, $ab \cdot af \cdot bf$ and $cd \cdot ce \cdot de$ respectively. The order of the letters is always the alphabetical one, viz. the single letters and duads denote pentads and double triads, thus:

$$\begin{aligned} a &= bcdef, & ab &= abf \cdot cde, \\ b &= acdef, & ac &= acf \cdot bde, \\ c &= abdef, & ad &= adf \cdot bce, \\ d &= abcef, & ae &= aef \cdot bcd, \\ e &= abcdf, & bc &= bcf \cdot ade, \\ f &= abcde, & bd &= bdf \cdot ace, \\ & & be &= bef \cdot acd, \\ & & cd &= cdf \cdot abe, \\ & & ce &= cef \cdot abd, \\ & & de &= def \cdot abc. \end{aligned}$$

There is no fear of ambiguity in writing (and we accordingly write) the squares of $\sqrt[4]{a}$ and $\sqrt[4]{ab}$ as $\sqrt{a}$ and $\sqrt{ab}$ respectively; the fourth powers are written $(\sqrt[4]{a})^2$ and $(\sqrt[4]{ab})^2$; the double stroke of the radical symbol $\sqrt$ is in every case perfectly distinctive.

This being so we have as above $c_0 = \lambda \sqrt[4]{bd}$, &c., $k_5 = \lambda \sqrt[4]{c}$, &c.: it is, however, important to notice that the fourth roots in question do not denote positive values, but they are fourth roots each taken with its proper sign (+, −, +i, −i, as the case may be) so as to satisfy the identical relations which exist between the sixteen constants; and it is not easy to determine the signs.

The variables x, y are connected with u, v by the differential relations

$$\begin{aligned} \varpi du + \tau dv &= -\frac{1}{2} \left(\frac{dx}{\sqrt{X}} - \frac{dy}{\sqrt{Y}} \right), \\ \sigma du + \rho dv &= -\frac{1}{2} \left(\frac{x dx}{\sqrt{X}} - \frac{y dy}{\sqrt{Y}} \right), \end{aligned}$$

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where $X = abcdef$, $Y = a_1 b_1 c_1 d_1 e_1 f_1$; which equations contain the constants $\varpi, \rho, \sigma, \tau$, the values of which will be afterwards connected with the other constants.

78. The c 's are expressed as functions of four quantities $\alpha, \beta, \gamma, \delta$, and connected with each other, and with the constants a, b, c, d, e, f , by the formulæ

$$\begin{aligned} 0 &= \alpha^2 + \beta^2 + \gamma^2 + \delta^2 = \omega_0'' \sqrt[4]{bd}, \\ 1 &= 2(\alpha\beta + \gamma\delta) = \omega_0'' \sqrt[4]{ce}, \\ 2 &= 2(\alpha\gamma + \beta\delta) = \omega_0'' \sqrt[4]{cd}, \\ 3 &= 2(\alpha\delta + \beta\gamma) = \omega_0'' \sqrt[4]{be}, \\ 4 &= \alpha^2 - \beta^2 + \gamma^2 - \delta^2 = \omega_0'' \sqrt[4]{ac}, \\ 6 &= 2(\alpha\beta - \gamma\delta) = \omega_0'' \sqrt[4]{ab}, \\ 8 &= \alpha^2 + \beta^2 - \gamma^2 - \delta^2 = \omega_0'' \sqrt[4]{bc}, \\ 9 &= 2(\alpha\gamma - \beta\delta) = \omega_0'' \sqrt[4]{de}, \\ 15 &= 2(\alpha\delta - \beta\gamma) = \omega_0'' \sqrt[4]{ae}. \end{aligned}$$

It hence appears that we can form an arrangement

$$\begin{vmatrix} c_4, & c_6, & c_1 \\ c_8, & -c_2, & c_9 \\ c_{15}, & -c_3, & -c_5 \end{vmatrix}, \quad \begin{vmatrix} a, & b, & c \\ a', & b', & c' \\ a'', & b'', & c'' \end{vmatrix},$$

a system of coefficients in the transformation between two sets of rectangular coordinates.

We have, between the squares of these coefficients of transformation, a system of 6 + 9 equations

$$\begin{aligned} a^2 + b^2 + c^2 &= 1, \\ a'^2 + b'^2 + c'^2 &= 1, \\ a''^2 + b''^2 + c''^2 &= 1, \\ a^2 + a'^2 + a''^2 &= 1, \\ b^2 + b'^2 + b''^2 &= 1, \\ c^2 + c'^2 + c''^2 &= 1, \end{aligned}$$

$$\begin{aligned} b^2 + c^2 &= a'^2 + a''^2, & b'^2 + c'^2 &= a''^2 + a^2, & b''^2 + c''^2 &= a^2 + a'^2, \\ c^2 + a^2 &= b'^2 + b''^2, & c'^2 + a'^2 &= b''^2 + b^2, & c''^2 + a''^2 &= b^2 + b'^2, \\ a^2 + b^2 &= c'^2 + c''^2, & a'^2 + b'^2 &= c''^2 + c^2, & a''^2 + b''^2 &= c^2 + c'^2. \end{aligned}$$

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that is,

$$c^4 + c^4 + c^4 - c^4 = 0;$$

12	+	1	+	6	−	0
9	+	4	+	3	−	0
2	+	15	+	8	−	0
12	+	9	+	2	−	0
1	+	4	+	15	−	0
6	+	3	+	8	−	0
1	+	6	−	9	−	2
6	+	12	−	4	−	15
12	+	1	−	3	−	8
4	+	3	−	2	−	12
3	+	9	−	15	−	1
9	+	4	−	8	−	6
15	+	8	−	12	−	9
8	+	2	−	1	−	4
2	+	15	−	6	−	3

and we have, between their products, a system of 6 + 9 equations

$$\begin{aligned} a'a'' + b'b'' + c'c'' &= 0, \\ a''a + b''b + c''c &= 0, \\ aa' + bb' + cc' &= 0, \\ bc + b'c' + b''c'' &= 0, \\ ca + c'a' + c''a'' &= 0, \\ ab + a'b' + a''b'' &= 0, \end{aligned}$$

$$\begin{aligned}
a, b, c &= b'c'' - b''c', & c'a'' - c''a', & & a'b'' - a''b', \\
a', b', c' &= b''c - bc'', & c''a - ca'', & & ab'' - a''b, \\
a'', b'', c'' &= bc' - b'c, & ca' - c'a, & & ab' - a'b.
\end{aligned}$$

C. X.

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